Impact of State of Charge on the Dynamic Hosting Capacity of Electric Vehicles in Radial Distribution Networks: An Experimentally Validated Study Toward Machine Learning-Based Generalization

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Abstract*—The active integration of electric vehicles (EVs) into distribution networks requires an accurate understanding of how the state of charge (SOC) of the connecting fleet affects hosting capacity (HC). Most existing HC studies treat EV charging as a constant-power or voltage-only load, and few link the SOC-dependent charging trajectory of a real vehicle to the resulting network-level hosting capacity. This paper develops and experimentally validates an SOC-dependent EV load model and re-evaluates dynamic hosting capacity as an explicit function of fleet SOC on the IEEE 33-bus radial distribution system. Three parameters that jointly govern the SOC-to-HC relationship are identified: battery open-circuit voltage, active charging power, and charger power factor. Battery open-circuit voltage and internal resistance are obtained from a CALCE incremental-current open-circuit-voltage (OCV) test on an INR 18650-20R cell and fitted with a five-term modified Nernst–Shepherd equation ($R^2 = 0.9934$). The resulting charging-power model is validated against an independently measured Opel Corsa-e charging session, reproducing the measured taper knee to within 2.6 percentage points. The reactive-power behaviour of the charger is characterised from over 94,000 measured samples, revealing a dominant load-independent component that collapses the fleet power factor from 0.982 to 0.753 near full charge. A closed-form hosting-capacity equation, $HC(s) = K/(P_{EV}(s) + \lambda Q_{EV}(s))$, is derived from the DistFlow voltage-drop relation and fitted to full power-flow simulation with $R^2 \approx 1$. Results show that hosting capacity, expressed in vehicle count, rises by a factor of 3.59 (no voltage support) to 2.76 (with on-load tap-changer support) from low to high fleet SOC, while hosting capacity expressed in active power falls by up to 46% over the same range because of the power-factor collapse. Consequently, assuming a fully charged fleet overestimates hosting capacity by roughly a factor of three, since vehicles arrive to charge precisely because their SOC is low. The result is shown to be robust to the treatment of voltage-dependent EV load behaviour.

***Index Terms*—Hosting capacity, electric vehicle charging, state of charge, distribution network, power factor, curve fitting, experimental validation, machine learning.**

I. Introduction

THE increasing penetration of electric vehicles (EVs) into low- and medium-voltage distribution networks is reshaping the load characteristics that network planners must account for. Hosting capacity (HC) analysis – the determination of how much new load or generation a network can accommodate without violating technical constraints – has consequently become a central tool for distribution system operators (DSOs) planning for EV uptake [1].

A substantial body of literature treats EV charging load as either a constant-power load (CPL) or, at best, a voltage-dependent ZIP load with parameters fitted from simulated charger behaviour [2]. Fewer studies explicitly model the dependence of instantaneous charging power on the state of charge (SOC) of the connecting vehicle, and fewer still validate that dependence against a real, measured charging session. Where SOC is considered, it is frequently introduced as a stochastic input sampled from a bounded distribution for Monte Carlo hosting-capacity assessment [3], rather than as an explicit, physically motivated driver of the charging-power trajectory itself.

This gap matters because the relationship between SOC and hosting capacity is not intuitive. A vehicle at low SOC draws close to its rated charging power throughout a constant-current (CC) phase; a vehicle near full charge tapers into a constant-voltage (CV) phase during which active power falls sharply while the charger's reactive draw does not fall proportionally. The net effect on network-level hosting capacity, measured either in the number of vehicles that can be accommodated or in the aggregate power that can be delivered, is therefore not obvious *a priori* and depends on which network constraint binds first.

This paper addresses that gap directly. Three parameters that lie at the intersection of "what hosting capacity depends on" and "what SOC influences" are identified: battery open-circuit voltage V_{oc}, active charging power P_{EV}, and charger power factor pf. Each parameter is obtained from measured data rather than assumed, propagated through a validated charging model, and used to re-evaluate hosting capacity as an explicit function of fleet SOC on the IEEE 33-bus radial test system.

The main contributions of this paper are:

- An SOC-dependent open-circuit-voltage and internal-resistance model fitted to a CALCE incremental-current OCV test, rather than to a synthetic or assumed battery curve.
- A charging-power taper model validated against an independently measured EV charging session, with the taper knee reproduced to within 2.6 percentage points.
- An experimentally derived reactive-power model showing that the charger's fixed reactive draw, not its active power,

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TABLE I
THE THREE PARAMETERS LINKING SOC TO HOSTING CAPACITY

Parameter	Role in Hosting Capacity	Influence of SOC
V_{oc}, battery open-circuit voltage	Sets the charger's operating point (CC vs. CV)	Rises directly and non-linearly with SOC
P_{EV}, active charging power	Drives voltage drop, thermal loading, and network losses	Falls sharply once SOC passes the taper knee
pf, charger power factor	A recognised hosting-capacity-limiting index [1]	Active power tapers; reactive draw does not

ultimately limits hosting capacity expressed in kilowatts.

- A closed-form equation for dynamic hosting capacity as a function of fleet SOC, motivated by the DistFlow voltage-drop relation and validated against full power-flow simulation with $R^2 \approx 1$.
- A demonstration that the principal SOC–HC relationship is robust to the treatment of voltage-dependent EV load behaviour, using literature-published ZIP coefficients [2].

The remainder of this paper is organised as follows. Section II describes the network model, the three linking parameters, and the experimental datasets used to obtain each. Section III presents the closed-form hosting-capacity formulation. Section IV presents the results, including the principal SOC–HC relationship, the power-factor mechanism, constraint analysis, and robustness checks. Section V discusses the scope and limitations of the study, and Section VI concludes.

II. Methodology

A. Test System and Load Flow

The IEEE 33-bus radial distribution feeder (12.66 kV, 100 MVA base) [4] is used throughout, carrying a base load of 3715 kW and 2300 kvar. Power flow is solved using the Backward/Forward Sweep (BFS) method, vectorised with a Bus-Injection-to-Branch-Current (BIBC) path matrix so that each solution requires approximately 0.12 ms, which is necessary because hosting-capacity determination requires thousands of repeated power-flow evaluations inside a bisection search. The base-case solution was validated against the published benchmark, giving a total active loss of 202.68 kW and a minimum voltage of 0.9131 pu at bus 18, matching the literature values to four significant figures [4].

B. The Three SOC–HC Linking Parameters

No direct analytical relationship connects SOC, an internal battery state, to hosting capacity, a network-level limit. Instead, three parameters are identified that (i) hosting capacity is known to depend on, and (ii) SOC is known to influence:

The causal chain linking these parameters is: as SOC rises, battery voltage rises (Section II-C); the charger is consequently forced from constant-current into constant-voltage operation, so charging current and active power fall (Section II-D); the converter's largely fixed reactive draw does not fall proportionally, so power factor collapses (Section II-E); and hosting capacity is affected through both the active- and reactive-power channels (Section IV).

C. Battery Open-Circuit Voltage and Internal Resistance

Open-circuit voltage and internal resistance were obtained from a CALCE (Center for Advanced Life Cycle Engineering, University of Maryland) incremental-current OCV test on an INR 18650-20R cell (LiNiMnCo/graphite, 2000 mAh nominal) at 25°C. The test protocol applies 720 s of 0.5C current per 10% SOC increment, followed by a 7200 s (2-hour) rest; the fully relaxed voltage at the end of each rest is taken as the OCV. Both charge and discharge directions were used, yielding 19 measured OCV–SOC points spanning the full SOC range, subsequently averaged to remove hysteresis (mean 7.6 mV, maximum 18.1 mV).

The measured cell values were scaled to a 96-series, 84-parallel (96S84P) pack configuration, giving a 59.72 kWh pack with an OCV range of 313.4–400.9 V. The pack OCV was fitted with a five-term modified Nernst–Shepherd equation:

$$V_{oc}(s) = a_0 + a_1 s + a_2 \ln s + a_3 \ln(1-s) + a_4 e^{a_5(s-1)} \quad (1)$$

where $s \in [0, 1]$ is the fractional SOC, clipped to $[0.002, 0.998]$ to avoid the logarithmic singularities. The fitted coefficients are $a_0 = 343.4252$, $a_1 = 8.8594$, $a_2 = 6.4001$, $a_3 = 1.3730$, $a_4 = 56.1039$, $a_5 = 3.0137$ (all in volts except a_5), giving $R^2 = 0.9934$ against the 19 measured points (RMSE $= 1.72$ V, or 17.9 mV per cell).

Internal resistance was extracted independently by two methods from the same dataset: the 1-second voltage step following each rest (mean 0.2282 Ω per cell), and the sustained voltage drop across each full 720-s current step (mean 0.2583 Ω per cell). The two agree to within 13%, confirming the measurement reflects a genuine cell property. Because charging is a sustained process, the sustained value is adopted and fitted linearly in SOC:

$$R_{cell}(s) = 0.26025 - 0.00411\, s \ [\Omega], \qquad R_{pack}(s) = \tfrac{96}{84} R_{cell}(s) \quad (2)$$

Notably, the measured resistance is essentially flat, with a slightly *negative* slope with SOC over the measured range (0.09–0.80), which does not support the commonly assumed monotonically rising trend.

D. Charging-Power Model and Experimental Validation

Driven by (1) and (2), a constant-current/constant-voltage (CC-CV) charging simulation was run to generate the SOC-dependent charging power, fitted with a generalised (Richards) logistic form:

$$P_{EV}(s) = P_{rated} \left[\frac{A}{\left(1 + e^{(s-s_0)/k}\right)^m} \right] \quad (3)$$

with fitted parameters $A = 1.0063$, $s_0 = 0.9500$, $k = 0.0200$, $m = 0.6330$ ($R^2 = 0.9320$), where s_0 is the taper knee – the SOC at which charging power begins to fall.

This model was validated against an independently measured charging session of an Opel Corsa-e (DC, 10 kW setpoint, 4.83 h, SOC 3–100%, 33,780 valid samples after cleaning). The measured session shows charging power essentially constant at 9.42 kW up to approximately 96% SOC, followed by a step-wise decline. Fitting the same functional form (3) to the measured curve gives an independent taper-knee estimate of $s_0 = 0.9757$, so that the model knee of 0.9500 matches the measured knee to within 2.6 percentage points – the central experimental validation of this study.

E. Reactive Power and Power Factor

The charger's reactive-power behaviour was characterised by regressing measured grid-side reactive power against active power over 94,024 samples from the Opel measurement campaign, with AC and DC charging treated separately (pooling gives $R^2 = 0.065$, indicating the two must be modelled independently):

$$Q_{EV}(s) = P_{EV}(s) \tan \phi_L + Q_{idle} \tag{4}$$

For DC charging (the case adopted here), $\tan \phi_L = 0.0281$ and $Q_{idle} = 1.664$ kvar ($R^2 = 0.719$, $n = 49{,}523$), with the fixed component Q_{idle} accounting for 87.3% of the mean reactive draw. Power factor follows algebraically as $\mathrm{pf}(s) = P_{EV}(s)/\sqrt{P_{EV}(s)^2 + Q_{EV}(s)^2}$, falling from 0.9818 at low SOC to 0.7529 at full charge.

F. Voltage Dependence of EV Load

The voltage sensitivity of the EV charging load is represented by a ZIP model,

$$P_{EV}(V,s) = P_{EV}(s) \left[Z_p \left(\frac{V}{V_0} \right)^2 + I_p \left(\frac{V}{V_0} \right) + P_p \right] \tag{5}$$

using coefficients published by Shukla *et al.* [2] ($Z_p = -0.021902$, $I_p = 0.032173$, $P_p = 0.989728$, satisfying $Z_p + I_p + P_p = 1$), rather than an assumed set. The negative Z_p correctly reproduces the source's finding that active power falls as supply voltage rises; the coefficient spread across SOC (of order 10^{-4}) confirms the envelope is SOC-invariant, so that SOC enters only through the magnitude $P_{EV}(s)$, not through the voltage-sensitivity shape.

G. Hosting-Capacity Constraint Set

Hosting capacity is defined as

$$HC_{DHC}(s) = \min \left(HC_V,\ HC_{th},\ HC_{tx},\ HC_{loss} \right) \tag{6}$$

with a voltage constraint $V \geq 0.90$ pu (calibrated to the base-case minimum of 0.9131 pu, since the standard 0.95 pu limit is already violated with zero EVs connected), a thermal constraint of 319 A (7 MVA at 12.66 kV), a transformer constraint of 8 MVA, and a network loss-rate constraint of 7%. Hosting capacity is located by bisection search on the *number of connected vehicles* rather than on total power, which both matches the practical planning question and ensures the reactive-power effect of Section II-E is captured per-charger rather than being absorbed into a power-based rescaling.

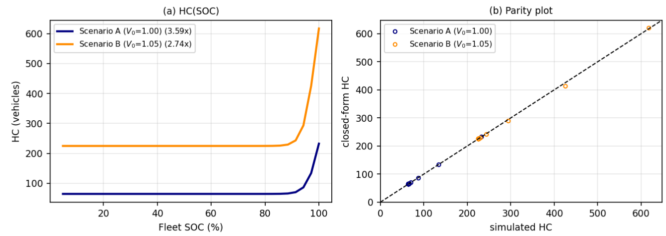

Fig. 1. Hosting capacity against fleet SOC for both network scenarios (left), and the closed-form equation (7) against full simulation (right).

TABLE II
HOSTING CAPACITY VS. FLEET SOC, SCENARIO A ($V_0 = 1.00$ PU)

SOC	P_0 (kW)	HC (vehicles)	HC (kW)	pf
5%	10.063	62.65	630.4	0.9818
50%	10.063	62.65	630.4	0.9818
80%	10.059	62.67	630.4	0.9818
90%	9.573	65.51	627.1	0.9802
100%	1.967	224.90	442.4	0.7529

III. CLOSED-FORM HOSTING-CAPACITY EQUATION

From the DistFlow voltage-drop approximation $\Delta V \approx (RP + XQ)/V$, the stress imposed by one additional vehicle is a weighted sum of its active and reactive power rather than its apparent power alone. This motivates the closed form

$$HC(s) = \frac{K}{P_{EV}(s) + \lambda\, Q_{EV}(s)} \tag{7}$$

where K (kW) represents the network's SOC-independent absorption headroom and λ is an effective X/R-like weighting of reactive against active power. Fitting (7) against full bisection-search hosting-capacity results across the SOC range gives, for a network with no substation voltage support (Scenario A, $V_0 = 1.00$ pu), $K = 717.08$ kW, $\lambda = 0.7104$, $R^2 = 1.00000$; and with on-load tap-changer (OLTC) support (Scenario B, $V_0 = 1.05$ pu), $K = 2903.68$ kW, $\lambda = 1.6349$, $R^2 = 0.99924$. The near-unity R^2 in both cases confirms that (7) reproduces the full simulation to a high degree of accuracy without requiring repeated power-flow evaluation once K and λ are known for a given network.

IV. RESULTS

A. Principal Result: Hosting Capacity vs. Fleet SOC

Table II and Fig. 1 summarise the principal result. Hosting capacity is nearly flat from 5% to approximately 80% SOC, since the charger remains in constant-current mode at full power throughout that range; the increase is concentrated beyond the taper knee at $s_0 = 0.950$.

The highest number of EVs is accommodated at high SOC. Moving from 5% to 100% SOC increases hosting capacity by a factor of 3.59 in Scenario A and 2.76 in Scenario B. Consequently, assuming a fully charged fleet overestimates hosting capacity by roughly a factor of three, since vehicles in practice arrive to charge precisely because their SOC is low; the realistic and conservative planning assumption is therefore a low-SOC fleet.

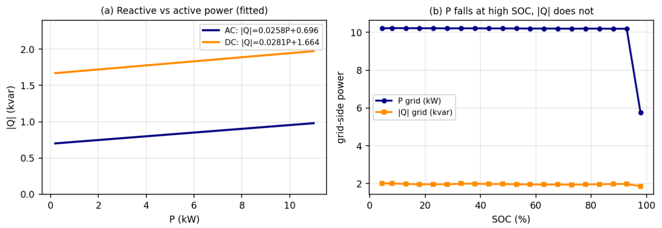

Fig. 2. Measured reactive power against active power for AC and DC charging (left), and the divergence of active and reactive grid-side power with SOC (right).

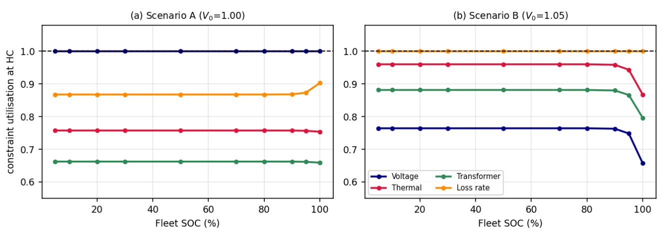

Fig. 3. Constraint utilisation at the hosting-capacity point for both scenarios.

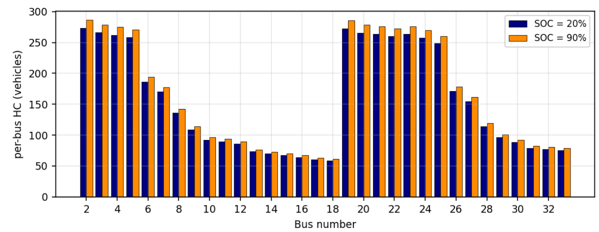

Fig. 4. Per-bus hosting capacity at two SOC levels, Scenario B.

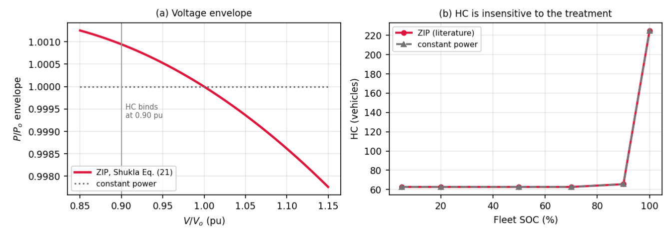

Fig. 5. Voltage envelope under the ZIP model of (5) (left), and hosting capacity under the ZIP model versus a pure constant-power model (right).

B. The Power-Factor Mechanism

Hosting capacity expressed in vehicle count and in active power move in *opposite* directions: while count rises 3.59×, hosting capacity in kilowatts *falls* by 29.8% (Scenario A) to 46.0% (Scenario B) over the same SOC range. Near full charge, a vehicle draws 1.97 kW of active power but 2.72 kvar of reactive power (Fig. 2), behaving close to a pure reactive load. This establishes that the SOC–HC relationship is physically grounded rather than a trivial rescaling: were hosting capacity in kilowatts SOC-independent, the vehicle-count result would follow from division alone.

C. Model-Free Cross-Check

Driving the hosting-capacity engine directly from the raw measured Opel charging curve, bypassing the fitted model of (3) entirely, gives results within 0.6% of the fitted-model results up to 80% SOC and within 4% at 90% SOC, confirming that the fitted closed form is not an artefact of curve-fitting but tracks the underlying measurement.

D. Constraint Analysis

Without voltage support, the voltage constraint binds throughout the SOC range, since the base-case minimum voltage (0.9131 pu) leaves only 0.0131 pu of headroom above the 0.90 pu limit. With OLTC support, the binding constraint shifts to network loss rate, with voltage utilisation falling to approximately 0.77. In this scenario, a voltage-only hosting-capacity assessment would overestimate HC by roughly a factor of four, underscoring the value of the composite multi-constraint formulation.

E. Per-Bus Hosting Capacity

Per-bus hosting capacity (Fig. 4) is lowest at bus 18, at the electrical end of the feeder (56.9 vehicles at 90% SOC), and highest at bus 2, nearest the substation (278.7 vehicles). The bus ranking is stable across SOC levels, while the spread between buses widens considerably at high SOC, providing a directly usable siting map for charging-station placement.

F. Robustness to the Voltage-Dependent Load Model

With $P_p = 0.9897$, the published ZIP model of (5) is within 0.1% of a pure constant-power load, and the SOC hosting-capacity multiplier is unchanged to three decimal places between the two treatments (Fig. 5). **The principal SOC–HC relationship is therefore robust to the treatment of voltage-dependent EV load behaviour.**

G. Fleet SOC Distribution

With the validated taper knee at $s_0 = 0.950$, active charging power is nearly constant across most of the SOC range, so that the spread of arrival SOC across a heterogeneous fleet has little effect on aggregate hosting capacity (less than 1% change in the Jensen gap between $E[P_{EV}(s)]$ and $P_{EV}(E[s])$ at the widest tested spread). For the charger class studied here, a mean-SOC representation of the fleet is therefore adequate.

V. Scope and Limitations

Results apply to an approximately 10 kW charger, the rating at which every model parameter – OCV, resistance, taper knee, and reactive-power coefficients – was experimentally measured; fast-charging behaviour at 0.5–1C is outside the validated scope. The charging-power validation of Section II-D rests on one complete measured session of one vehicle; a second session, ideally of a different vehicle model, would strengthen the generality of the result. Battery parameters derive from a single cell sample at 25°C, with resistance measured over the SOC range 0.09–0.80; behaviour outside this range or at other temperatures is extrapolated under the assumption of a balanced, isothermal pack. The voltage limit of 0.90 pu is calibrated to this network's own base-case minimum voltage and should be re-derived for other networks. Vehicle-to-grid operation is outside the present scope, which considers grid-to-vehicle charging only.

VI. Future Directions: Machine Learning-Based Generalization

The closed-form model of (7) is fitted to a single network topology, a single charger rating, and fixed constraint limits; its coefficients K and λ must be re-derived by repeated bisection-search power-flow evaluation whenever the topology, EV population, or constraint set changes. This is precisely the computational bottleneck that machine learning (ML) is well suited to remove, and it defines the natural next stage of this work.

A. HC Regression Across Stochastic Scenarios

Following the two-stage approach of [3], a dataset of stochastic scenarios – varying EV count, connection location, fleet SOC distribution parameters (mean and spread), and substation voltage support – can be generated by repeated power-flow evaluation using the validated models of Section II, and used to train a regression model (e.g., an artificial neural network) that predicts hosting capacity directly from these features. This would generalize the present single-scenario closed form of (7) to arbitrary combinations of stochastic EV behavior without requiring repeated bisection search for every new scenario, providing a practical, near-instantaneous decision-support tool for a DSO.

B. Operational-Suitability Classification

A classification stage, trained to predict whether a given stochastic scenario satisfies all voltage, thermal, transformer, and loss-rate constraints before a regression stage estimates achievable hosting capacity, would mirror the classification-then-regression pipeline of [3] and allow rapid triage of connection requests without full power-flow evaluation.

C. Feature Importance for Model Refinement

Explainability techniques such as SHapley Additive exPlanations (SHAP) could quantify which stochastic parameters – fleet SOC distribution, EV count, EV location, or charger rating – most strongly drive near-limit constraint violations. This would directly target the principal open assumption identified in Section V: whether the fixed reactive-power component Q_{idle} scales with converter rating. Training on a dataset spanning multiple measured charger ratings would allow this scaling law to be learned rather than assumed, resolving the single largest source of uncertainty carried into (7).

D. Generalized Charging-Power and Power-Factor Mapping

Finally, an ML regression trained on charging sessions across multiple vehicle models and charger ratings could learn a generalized SOC-to-power-factor mapping, replacing the single-vehicle, single-rating model of Section II-D and II-E with a data-driven function of charger rating and vehicle class, extending the experimental validation of this study beyond the one measured session on which it currently rests (Section V).

VII. Conclusion

This paper developed and experimentally validated an SOC-dependent EV charging-load model and used it to re-evaluate the dynamic hosting capacity of the IEEE 33-bus distribution network as an explicit function of fleet state of charge. Three linking parameters – battery open-circuit voltage, active charging power, and charger power factor – were each obtained from measured data: a CALCE incremental-current OCV test for the battery parameters, and an independently measured Opel Corsa-e charging session for the charging-power taper and reactive-power behaviour, with the model's taper knee validated to within 2.6 percentage points of the measured value. A closed-form hosting-capacity equation, motivated by the DistFlow voltage-drop relation, reproduced full power-flow simulation results with $R^2 \approx 1$.

The central finding is that hosting capacity, in vehicle count, rises by a factor of 2.76 to 3.59 from low to high fleet SOC, while hosting capacity in active power *falls* by up to 46% over the same range because of a collapse in charger power factor near full charge. The practical implication for distribution planning is that assuming a fully charged EV fleet substantially overestimates hosting capacity; the low-SOC condition, which is also the condition in which vehicles realistically arrive to charge, is the binding and conservative case. This result was shown to hold regardless of whether EV load voltage-dependence is modelled with literature-published ZIP coefficients or as pure constant power, indicating that the SOC-driven charging-power and power-factor mechanisms, not the voltage-sensitivity treatment, are what govern the outcome.

Future work should extend the experimental validation to a higher-C-rate (fast) charging session, a second measured vehicle, and multiple battery temperatures, and should examine whether the fixed reactive-power component identified here scales with converter rating, which remains the principal untested assumption carried into the closed-form equation.

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